1-(MAT/12_HL_Summer_2012_Q4)-*BinomialTheorem*

(a) Expand and simplify $\left(x-\frac{2}{x}\right)^4$. *[3 marks]*

(b) Hence determine the constant term in the expansion $(2x^2+1)\left(x-\frac{2}{x}\right)^4$. *[2 marks]*

2-(MAT/10_HL_Winter_2012_Q2)**-***BinomialTheorem*

[Maximum mark: 4]

Expand and simplify $\left(\frac{x}{y}-\frac{y}{x}\right)^4$.

3-(MAT/12_HL_Summer_2013_Q3)**-***BinomialTheorem*

[Maximum mark: 4]

Expand $(2-3x)^5$ in ascending powers of x, simplifying coefficients.

4–(MAT/11_HL_Summer_2013_Q13)–*Probability,BinomialTheorem*

On Saturday, Alfred and Beatrice play 6 different games against each other. In each game, one of the two wins. The probability that Alfred wins any one of these games is $\frac{2}{3}$.

(a) Show that the probability that Alfred wins exactly 4 of the games is $\frac{80}{243}$. *[3 marks]*

(b) (i) Explain why the total number of possible outcomes for the results of the 6 games is 64.

(ii) By expanding $(1+x)^6$ and choosing a suitable value for x, prove

$$64=\binom{6}{0}+\binom{6}{1}+\binom{6}{2}+\binom{6}{3}+\binom{6}{4}+\binom{6}{5}+\binom{6}{6}.$$

(iii) State the meaning of this equality in the context of the 6 games played. *[4 marks]*

(c) The following day Alfred and Beatrice play the 6 games again. Assume that the probability that Alfred wins any one of these games is still $\frac{2}{3}$.

(i) Find an expression for the probability Alfred wins 4 games on the first day and 2 on the second day. Give your answer in the form $\binom{6}{r}^2\left(\frac{2}{3}\right)^s\left(\frac{1}{3}\right)^t$ where the values of r, s and t are to be found.

(ii) Using your answer to (c)(i) and 6 similar expressions write down the probability that Alfred wins a total of 6 games over the two days as the sum of 7 probabilities.

(iii) Hence prove that $\binom{12}{6}=\binom{6}{0}^2+\binom{6}{1}^2+\binom{6}{2}^2+\binom{6}{3}^2+\binom{6}{4}^2+\binom{6}{5}^2+\binom{6}{6}^2$. *[9 marks]*

(d) Alfred and Beatrice play n games. Let A denote the number of games Alfred wins. The expected value of A can be written as $\mathrm{E}(A)=\sum_{r=0}^{n} r\binom{n}{r}\frac{a^r}{b^n}$.

(i) Find the values of a and b.

(ii) By differentiating the expansion of $(1+x)^n$, prove that the expected number of games Alfred wins is $\frac{2n}{3}$. *[6 marks]*

5-(MAT/12_HL_Summer_2015_Q2)**-***BinomialTheorem*

[Maximum mark: 4]

Expand $(3-x)^4$ in ascending powers of x and simplify your answer.

6-(MAT/11_HL_Summer_2015_Q4)-*BinomialTheorem,Differentiation*

(a) Expand $(x+h)^3$. [2]

(b) Hence find the derivative of $f(x)=x^3$ from first principles. [3]

7–(MAT/10_HL_Winter_2015_Q3)–*BinomialTheorem*

(a) Write down and simplify the expansion of $(2+x)^4$ in ascending powers of x. [3]

(b) Hence find the exact value of $(2.1)^4$. [3]

8-(MAT/12_HL_Summer_2016_Q6)-*BinomialTheorem*

Consider the expansion of $(1+x)^n$ in ascending powers of x, where $n \geq 3$.

(a) Write down the first four terms of the expansion. [2]

The coefficients of the second, third and fourth terms of the expansion are consecutive terms of an arithmetic sequence.

(b) (i) Show that $n^3 - 9n^2 + 14n = 0$.

(ii) Hence find the value of n. [6]

9-(MAT/12_HL_Summer_2017_Q1)**-***BinomialTheorem*

[Maximum mark: 5]

Find the term independent of x in the binomial expansion of $\left(2x^2+\dfrac{1}{2x^3}\right)^{10}$.

10-(MAT/10_HL_Winter_2017_Q4)-*BinomialTheorem*

[Maximum mark: 4]

Find the coefficient of x^8 in the expansion of $\left(x^2 - \dfrac{2}{x}\right)^7$.

11-(MAT/11_HL_Summer_2019_Q2)-*BinomialTheorem*

[Maximum mark: 4]

Determine the first three terms of $(1-2x)^{11}$ in ascending powers of x, giving each term in its simplest form.

12-(MathAA/12_HL_Summer_2021_Q3)-*BinomialTheorem*

[Maximum mark: 5]

In the expansion of $(x+k)^7$, where $k \in \mathbb{R}$, the coefficient of the term in x^5 is 63.

Find the possible values of k.

13-(MathAA/10_HL_Winter_2021_Q9)-*BinomialTheorem*

[Maximum mark: 7]

Consider the expression $\dfrac{1}{\sqrt{1+ax}}-\sqrt{1-x}$ where $a \in \mathbb{Q}$, $a \neq 0$.

The binomial expansion of this expression, in ascending powers of x, as far as the term in x^2 is $4bx + bx^2$, where $b \in \mathbb{Q}$.

(a) Find the value of a and the value of b. [6]

(b) State the restriction which must be placed on x for this expansion to be valid. [1]

14-(MathAA/11_HL_Summer_2022_Q6)-*BinomialTheorem*

[Maximum mark: 5]

Consider the expansion of $\left(8x^3 - \dfrac{1}{2x}\right)^n$ where $n \in \mathbb{Z}^+$. Determine all possible values of n for which the expansion has a non-zero constant term.